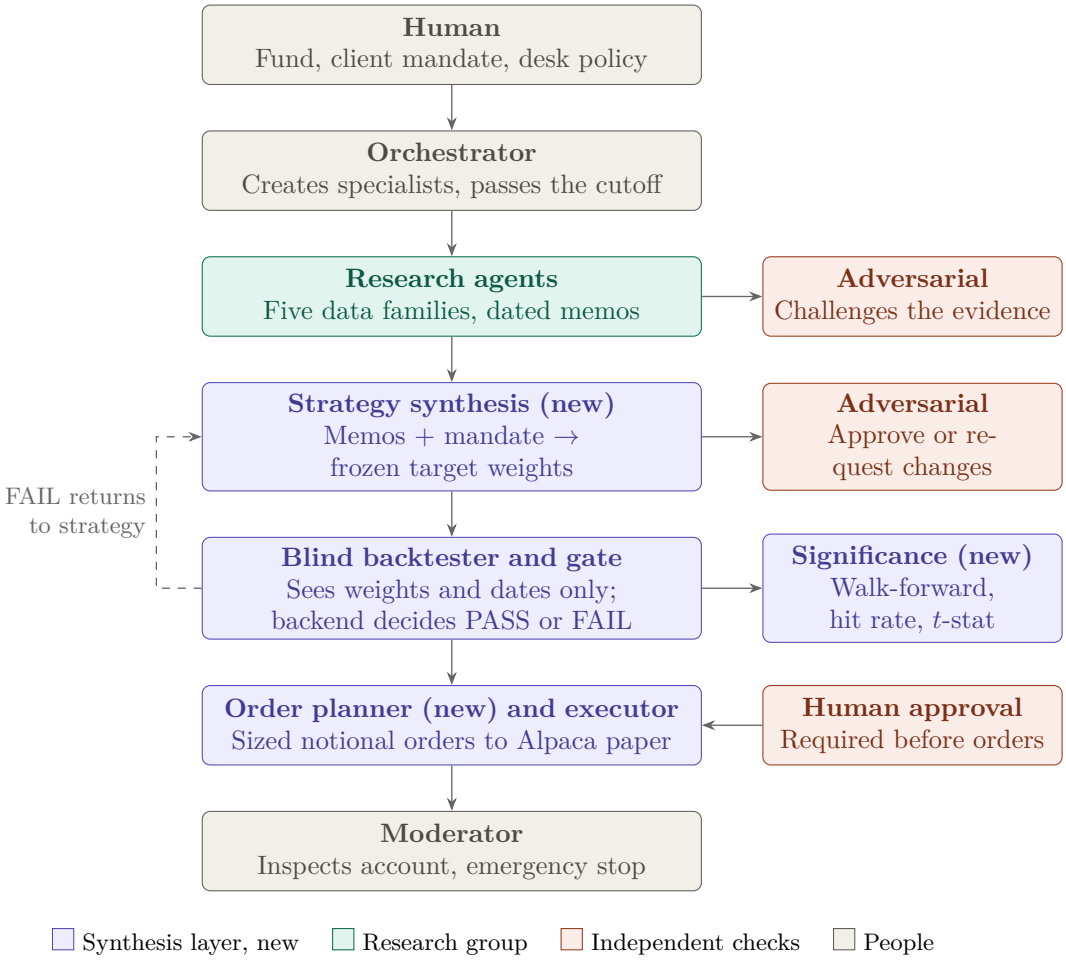
Agent flow and variables

Team 4, Track 3: Financial Portfolio Management

Austin Agentic AI Hackathon, 23 October 2026

Two views of the trading system as it exists on `main`, plus the synthesis layer added in `backend/app/synthesis/`. Roles and tools come from `backend/app/roles.py`, gates from `backend/app/desk.py`, execution from `backend/app/paper.py`. Boxes marked **new** are the synthesis-layer additions.

1. High-level flow



Gates the model cannot bypass. A strategy must be registered (frozen) before review. The backtester is created only after an approving review and sees nothing but the weights and dates. The backend computes PASS or FAIL from human thresholds. Execution rechecks permission, PASS, review, approval, limits and the stop before every order. A submitted order is not a fill.

2. Which variables each agent uses

Agent	Reads	Produces	Bound by
Research, live	Financial Datasets statements, ratios, earnings, prices, filing sections, insider and institutional ownership, company news; Exa web search; X search; FRED series; Jev sentiment; Apify	One memo per data family: ticker, stance (long, neutral, avoid), confidence 0 to 1, note, source ids with dates	Live data never enters a historical decision
Research, historical	Cutoff-filtered snapshots: OHLCV closes; SEC revenue, net income, total debt keyed on filing date; FRED CPI, FEDFUNDS, DGS10, UNRATE keyed on release date; curated events and GDELT tone	Same memo shape; every source dated on or before the cutoff	`assert_no_lookahead` on every cited source
Strategy synthesis (new)	All memos; mandate universe, capital, risk band, max_position, max_drawdown; policy universe and max_position	Frozen strategy: name, weights summing to 1, cutoff, end, rationale, memo paths, changes_my_mind. Seed is 1 plus signed convictions; net-avoid names dropped; clamped to max_position; optional single Bedrock refinement that is re-clamped in code	`register_strategy` rejects weights outside the universe or above max_position; revisions get new ids
Adversarial	The research or the registered strategy, plus native evidence tools	APPROVE or CHANGES_REQUESTED with a critique	Cannot review a strategy by its own author
Backtester	Only strategy id, weights, cutoff, end. The engine adds closes, SPY and the risk-free rate (FEDFUNDS at the cutoff)	total_return, cagr, max_drawdown, sharpe for portfolio and SPY	No file, web, research or delegation tools; bound to one strategy id
Gate	The metrics above; policy max_drawdown, min_sharpe, min_excess_return	PASS or FAIL with per-check results and a policy hash	Backend computes the verdict; a changed policy invalidates PASS
Significance (new)	Frozen weights; every monthly cutoff in range; fixed horizon	Number of windows, hit rate vs SPY and equal weight, mean excess return, t-statistic, p-value, bootstrap 95% interval	Reported beside the gate, never replaces it; windows overlap, so the t-stat is optimistic
Order planner (new)	Weights; capital as the smaller of account equity and mandate capital; account cash, positions, open orders; max_order_notional, max_daily_orders	Ordered list of buy and sell notional orders with stable client_order_ids, or blocked with a reason	Sells before buys; nothing above cash, position cap or notional limit

Agent	Reads	Produces	Bound by
Executor	The order plan; live web context only as a reason to defer	Broker order id, status, filled_qty, filled_avg_price	Stop clear, trading_allowed, current PASS, APPROVE, human approval and limits rechecked per order; an unknown result is never resubmitted
Moderator	Desk records, paper account, positions versus submitted orders	`emergency_stop` with a reason	Can engage the stop; only a person clears it

3. How this maps onto the roles on `main`

Kushagra's written role requirements take priority over the picture: the roster is flexible, one root can create more orchestrators and task-focused researchers, and the boxes above are context, not a fixed roster. The synthesis boxes are **tools on existing roles**, not new roles: `synthesize_strategy` on the strategy role, `plan_paper_orders` and `execute_order_plan` on the executor role. The blind backtester implementation is Mayank's (lane 2b); synthesis hands it a frozen strategy record (weights, cutoff, end) by `strategy_id` and consumes its PASS or FAIL. The significance report is an optional add-on and can be replaced by Mayank's walk-forward once it lands.

4. Limits to state in the demo

- Historical evaluation is the recorded monthly target-weight engine on cached closes: no dividends, no costs, fully invested. Significance is a walk-forward over cutoffs with the same frozen weights, which measures robustness of the allocation, not of a live rebalancing process.
- Live provider data is not point-in-time safe and never feeds a historical decision.
- Hiding research from the evaluator does not remove model-memory leakage; see Mayank's `docs/track3-plan.md` for the masked-ticker audit.
- Alpaca paper is a shared account; a submitted order is not a fill.